

CrewmateA310 — User Manual

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CrewmateA310 — User Manual

CrewmateA310 is a virtual First Officer companion for the **Airbus A310** in Microsoft Flight Simulator. It listens to your voice, responds with audio callouts, runs automated cockpit flows, and guides you through interactive checklists — just like a real crew member.

 Crewmate avatar

Getting Started

Requirements

- The EN-US voice package must be installed to use the voice engine.
- To use the trainer app, set your Display Language to EN-US while training (you can change it back afterward).

Voice Modes

CrewmateA310 supports two voice recognition modes:

Mode	How it works
Continuous	The microphone is always listening. Speak naturally.
Push-to-Talk (PTT)	Not implemented yet.

Volume & Voice Sensitivity

- **Sound Volume** — Controls how loud the FO's audio callouts are (0–100).
- **Voice Sensitivity** — Lower settings are more tolerant of variations (may increase false positives); higher settings are stricter.

Tutorial

Assumptions

This tutorial assumes you are parked at the gate with engines off. You are the Captain and PF (Pilot Flying); CrewMate acts as PM (Pilot Monitoring).

Typical preflight timeline (example)

- 45 min: Crew arrives at aircraft and PM takes their seat.
- 43 min: PM starts preliminary cockpit preparation.
- 42 min: PM departs for external walkaround. PF starts cockpit preparation on their side.

COCKPIT PREPARATION	
PF	PNF
GEAR PINS + COVERS CHECK OVHD PANEL	
* ALL WHITE LIGHTS EXTINGUISH	
* IRS NAV	
FMS LAT/LONG CHECK/CORRECT	
ALIGN IRS PROMPT PRESS	
* SIGNS SET	
* CALLS CHECK	
HYD quantities NORMAL RANGE	
* FLT RCDR GND CTL ON	
ISDU (if installed) OFF	
* EXT LTS AS REQ	
* PITCH TRIM/YAW DAMPER/ATS ON	
ENG 1 FIRE PROTECTION CHECK/TEST	
EVAC SIGNAL AS REQ	
L/G ANN CHECK	
* FUEL panel CHECK	
* FUEL X-FEED (if ER flight) TEST	
* CVR TEST	
CABIN PRESS CHECK	
* CREW OXY CHECK	
ENG 2 FIRE PROTECTION CHECK/TEST	
* EMER EXIT LT ARM	
COMPT TEMP CHECK/AS REQ	
* COM SET	
* AIRFIELD DATA OBTAIN	

* FMS INITIALIZATION
 DATA BASE VALIDITY CHECK
 NAVAID DESELECTION AS REQ
 F-PLN INITIALIZATION COMPLETE
 F-PLN A CHECK/COMPLETE
 F-PLN B CHECK
 SEC F-PLN AS REQ
 * FMS DATA INSERTION
 ZFW/CG or ZFCG/BLOCK FUEL INSERT
 T. O. DATA INSERT
 THR RED/ACC ALT SET/CHECK

. When both pilots seated :

GLARESHIELD :

* PFD/ND BRIGHTNESS AS REQ
 * FPV/FD ON FD
 * VOR/NAV/ILS AS REQ
 * EFIS cont panel AS REQ
 * ND mode/range AS REQ
 * FCU CHECK/SET

LATERAL CONSOLE :

RAT CHECK OFF
 OXYGEN MASK TEST
 GND SVS INTPH OFF

* PFD/ND BRIGHTNESS AS REQ
 * FPV/FD ON FD
 * VOR/NAV/ILS AS REQ
 * EFIS cont panel AS REQ
 * ND mode/range AS REQ
 * FCU CHECK/SET

RAT CHECK OFF
 OXYGEN MASK TEST

- 30 min: PM returns and starts cockpit preparation on their side.

R

CM1/CM2 INST PANELS :

GPWS NORM
 PFD/ND CHECK
 STBY ASI CHECK
 VSI CHECK
 ALTIMETER CHECK
 * BARO REF SET
 CLOCK CHECK/ADJUST

CTR INST PANEL

STBY ALT CHECK
 * BARO REF SET
 STBY HORIZON CHECK
 * ECAM RCL
 BRK-A/SKID NORM/ON
 * ENGINE INST CHECK
 * OIL QUANTITY CHECK
 * REV and REV UNLK lts CHECK
 * TRP CHECK
 L/G WARN TEST PRESS
 L/G POSITION DET SYS AS REQ
 LDG ELEV SET DEPARTURE FIELD ELEV

PFD/ND CHECK
 STBY ASI CHECK
 VSI CHECK
 ALTIMETER CHECK
 * BARO REF SET
 CLOCK CHECK/ADJUST

PEDESTAL : ADF CHECK * RADIOS CHECK * WX RADAR CHECK OFF * PARK BRK CHECK OFF/AS REQ * ATC TRANSPONDER STBY/OFF * TCAS (if installed) OFF VOR/ILS AS REQ * <u>FMS DATA CONFIRMATION</u> <u>FUEL</u> * <u>TAKE-OFF BRIEFING</u> PERFORM	IRS ALIGN CHECK INIT PAGE A/B CHECK T.O. DATA CALCULATE/CONFIRM F-PLN CHECK AIRFIELD DATA CONFIRM DFDR fault lights CHECK EXTINGUISHED * FUEL QUANTITY CHECK * FUEL PUMPS ON
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- 25 min: PF conducts the departure briefing (enter takeoff data in the Takeoff Performance window).
- 10 min: PF calls for the BEFORE START checklist to the line.
- 1 min: CrewMate closes the cockpit door. PF and PM perform before start flow, after that PF calls for the BEFORE START checklist below the line.

BEFORE PUSH BACK or START	
PF	PNF
LOADSHEET CHECK (CM1) T. O. DATA PREPARE/REVISE SEAT ADJUST CDU TAKE-OFF (PF) BEFORE START C/L DOWN TO THE LINE PUSHBACK/START CLEARANCE OBTAIN WINDOW/DOORS CLOSED PARK BRK AS REQ BEFORE START C/L BELOW THE LINE	T. O. DATA PREPARE/REVISE SEAT ADJUST CDU F-PLN (PNF) EXT PWR OFF/DISCONNECTED WINDOW CLOSED BEACON ON/AUTO

Before Start flow pattern

Pushback and Engine Start

- Command PM to start engines (e.g., "Start engine 2). PM will set ignition knob to Start A or B and press the starter.
- When ignition is set to OFF, PF and PM will start the AFTER START flow pattern. If anti-ice will be used, flaps may be left as required.
- On hand signal from ground personnel, call for the AFTER START checklist.

AFTER START	
PF	PNF
IGNITION OFF	APU BLEED OFF APU OFF
ECAM CHECK STS ECAM DOOR PAGE CHECK CLEAR To disconnect ANNOUNCE	ENG ANTI ICE AS REQ WING ANTI ICE AS REQ SLATS/FLAPS SET GND SPOILERS ARM AIL/RUD TRIM ZERO PITCH TRIM SET
AFTER START C/L	

After Start flow pattern

Taxi

- PM announces when the cabin is ready.
- Check flight controls at a convenient time before or during taxi (this is done before arming the autobrake).
- Control check sequence: Full Up, Full Down, Full Left, Full Right, Rudder Full Left, Rudder Full Right.
- After the controls check, PM performs the 2nd part of the TAXI flow pattern.
- After T.O. CONFIG pushbutton is pressed and a cabin report is received, PF calls for the BEFORE TAKEOFF to the line checklist.

TAXI	
PF	PNF
NOSE LIGHT TAXI	TAXI CLEARANCE OBTAIN
PARK BRK OFF	ELAPSED TIME START
EXT LTS AS REQ	
THRUST AS REQ	ANNOUNCE "PRESSURE ZERO"
BRAKES CHECK	FLT CTL CHECK
FLT CTL CHECK	ATC CLEARANCE OBTAIN/CONFIRM
T. O. DATA/CONDITIONS CHECK/REVISE	T.O. DATA/CONDITIONS CHECK/REVISE
	FMS F-PLN CHECK
	FCU ALT/HDG/SPD CHECK
	PROF ARM
	NAV ARM
FD CHECK ON	FD CHECK ON
FMA CHECK	FMA CHECK
FLT INST CHECK	FLT INST CHECK
	RADAR AS REQ
TAKE-OFF BRIEFING CONFIRM	ATC CODE CONFIRM/SET
CABIN REPORT	OBTAIN (CM1)
BEFORE TO C/L DOWN TO THE LINE	T.O. CONFIG TEST PRESS
	ECAM NORM FOR T. O.

Taxi flow pattern

Line-up & Takeoff

- PF calls for the Line-up flow.

- When line-up clearance is received and the Line-up flow pattern is complete, PF calls for the BEFORE TAKEOFF below the line checklist.
- When cleared for takeoff, announce "TAKEOFF."

BEFORE TAKE OFF	
PF	PNF
APPROACH PATH CLEAR OF TRAFFIC EXT LTS SET QFU & THRESHOLD CONFIRM BEFORE TO C/L BELOW THE LINE	TAKE OFF/LINE UP CLEARANCE OBTAIN CABIN CREW ADVISE BRAKE PRESSURE CHECK BRAKE TEMP (if BRK FAN on) CHECK BRAKE FAN (if installed) OFF AUTO BRK MAX IGNITION CONT RELIGHT/AS REQ PACK VALVE AS REQ ATC TRANSPONDER ON/ALT REP TCAS (if installed) TA/RA/AS REQ QFU & THRESHOLD CONFIRM

Line-up flow pattern

Acceleration

- The First part of the After-Takeoff flow is triggered when you are above thrust reduction altitude.
- The Second part of the After-Takeoff flow is triggered when the aircraft is clean.
- When the aircraft is clean call for the AFTER TAKEOFF CLIMB checklist to the line.
- When the aircraft is above transition altitude and altimeters are standard call for the AFTER TAKEOFF CLIMB checklist below the line.

TAKE OFF	
PF	PNF
ANNOUNCE "TAKE OFF" CLOCK START BRAKES RELEASE GO LEVERS TRIGGERS (CM1) CM1 keeps his hand on throttles until V1 is reached. PFD/ND SCAN FMA INDICATION ANNOUNCE • Below 80 Kts : • At 100 Kts : • At V1 : • At VR : • At VSI positive : ORDER "GEAR UP" AP ON/AS REQ • At thrust red alt FMA/TRP CHECK • At F speed ORDER "FLAPS ZERO" • At S speed ORDER "SLATS RETRACT"	CLOCK START PFD/ND SCAN N1 (EPR) CHECK THRUST SET ANNOUNCE AIRSPEED/ENG INST SCAN ANNOUNCE "100 KTS" ANNOUNCE "V1" ANNOUNCE "ROTATE" ANNOUNCE "POSITIVE CLIMB" L/G UP GND SPOILERS DISARM EXT LTS SET L/G LEVER NEUTRAL ANNOUNCE "GEAR UP/NEUTRAL" ONE PACK ON FLAPS 0 SELECT ANNOUNCE "FLAPS ZERO" SLATS 0 SELECT ANNOUNCE "SLATS RETRACTED" 2ND PACK ON

AFTER TAKE OFF	
PF	PNF
ANTI ICE AS REQ AFTER TO/CLIMB C/L DOWN TO THE LINE	APU BLEED AS REQ APU AS REQ IGNITION AS REQ FUEL CTR TK PUMPS ON SIGNS AS REQ TCAS (if installed) TA/RA

Climb to 10,000 ft

CLIMB	
PF	PNF
FMA MONITOR/ANNOUNCE ANY CHANGES PF CDU PROG FCU/FMS SET (IF AP ON) • At transition altitude ALTIMETER STD CRZ FL CHECK/MODIFY AFTER TO/CLIMB C/L BELOW THE LINE RADAR AS REQ	LAND LTS OFF/RETRACT FMA MONITOR ANY CHANGES PNF CDU F-PLN FCU/FMS SET (IF AP OFF) ALTIMETER STD RADAR AS REQ

Climb/10,000 ft flow pattern

Descent Preparation

- PF should insert landing data in the Landing Performance window.

DESCENT PREPARATION	
PF	PNF
	ECAM MEMO/STS CHECK
	WEATHER/LANDING INFO OBTAIN
	LANDING ELEV SET
	FUEL CHECK
	LANDING DATA PREPARE
FMS PREPARE	FMS PREPARE
APPROACH BRIEFING PERFORM	
	DESCENT CLEARANCE OBTAIN
FCU ALT CLEARED ALT	
	ANTI ICE AS REQ
	IGNITION AS REQ

Descent preparation items

10k Descent

INITIAL APPROACH	
PF	PNF
	IGNITION CONT RELIGHT/AS REQ
	SIGNS SET
	EXT LTS AS REQ
POSITIONING MONITOR	POSITIONING MONITOR
	NAV/COM CHECK/SET
ILS SELECT/AS REQ	
NAV ACCURACY CHECK	
RADAR TILT ADJUST	
TERR ON ND AS RQRD	TERR ON ND AS RQRD
ND RANGE 15NM	ND RANGE 15 NM
ND MAP FOR ILS (OTHERWISE ROSE OR ARC)	ND MAP/AS REQ
• For CAT II OR CAT III :	
DH CHECK/SET	DH CHECK/SET
APPROACH C/L	

10k descent flow pattern

Approach

- After passing the transition level and/or setting the barometric reference, complete the APPROACH checklist.

Landing

- When LDG CONF is set and a cabin report is received, call for the LANDING checklist.
- PF announces "Continue" at minima or "Go-around - flaps" as appropriate.

After Landing

- Disarming the ground spoilers will trigger the After-Landing flow.
- If anti-ice is used, flaps may remain at LDG CONF per procedure.
- Once flows done call for the AFTER LANDING checklist.

AFTER LANDING	
PF	PNF
LANDING LTS OFF/RETRACT	ICE PROTECTION AS REQ
GND SPOILERS DISARM	IGNITION OFF
	APU START
	TRANSPONDER STBY/OFF
	TCAS (if installed) STBY OFF
	RADAR OFF/TEST
	PITCH TRIM 1° NOSE UP
	SLATS/FLAPS RETRACT
	BRAKE TEMP CHECK
	BRAKE FANS (if installed) AS REQ
AFTER LANDING C/L	

After landing flow pattern

Parking

- Turn off taxi lights when turning into the gate, or ask the PM to confirm.
- Shutting down the engines will trigger the Parking flow.
- After the Parking flow pattern completes, PM calls for the PARKING checklist.

Voice Commands

Speak these phrases clearly during flight. The FO uses partial matching — you don't need to be word-perfect, but include the key phrase.

Preflight Timer

Say	What happens
"Let's prepare the aircraft"	Starts the preflight countdown timer to help you track preparation time.

Ground Engineer

Say	What happens
"Ground from cockpit"	Ground Engineer will ask what's your request
"Cockpit to ground"	Ground Engineer will ask what's your request

Gear

Say	What happens
"gear down"	Lowers the landing gear. Speed must be at or below 270 knots.
"gear up"	Raises the landing gear.

Flaps

The FO will confirm speed limits before moving flaps while airborne.

Say	Flap Setting	Max Speed
"slats retract"	Flaps 0/0 (retracted)	—
"slats extend" or "flaps 0"	Flaps 15/0	245 kts
"flaps fifteen"	Flaps 15/15	210 kts
"flaps twenty"	Flaps 20/20	195 kts
"flaps forty"	Flaps 40	180 kts

Engine Anti-Ice

Say	What happens
"Engine anti ice on"	Turns on engine anti-ice for both engines.
"Engine anti ice off"	Turns off engine anti-ice.

Wing Anti-Ice

Say	What happens
"Wing anti ice on"	Turns on wing anti-ice.
"Wing anti ice off"	Turns off wing anti-ice.

Lights (Landing / Taxi / Nose / Strobe)

Say	What happens
"Landing lights on"	Turns on landing lights.
"Landing lights off"	Turns off landing lights.
"Taxi lights on"	Turns on nose taxi light.
"Taxi lights off"	Turns off nose taxi light.
"Takeoff light on"	Turns on nose takeoff light.
"Strobe lights on"	Turns on strobes.
"Strobe lights auto"	Sets strobes to AUTO.
"Strobe lights off"	Turns off strobes.

Flight Director

Say	What happens
"Flight Director on"	Activates the Flight Director.
"Flight Director off"	Deactivates the Flight Director.

Autopilot

Say	What happens
"Autopilot on"	Engages Autopilot 1.
"Set speed <i>or speed select</i> "	Sets commanded speed.
"Set heading _ <i>or heading select</i> _"	Sets commanded heading.
"Set altitude _ or altitude select _ or set flight level <i>or flight level select</i> "	Sets commanded altitude.
"Pull speed"	Pulls speed knob to select selected speed.
"Pull speed _"	Pulls speed knob and sets the commanded speed.
"Pull heading"	Pulls heading knob to select selected heading.
"Pull heading _"	Pulls heading knob and sets the commanded heading.
"Heading push"	Pushes heading knob to synchronize FCU heading with aircraft heading
"Engage nav"	Selects NAV mode.
"Arm land"	Selects LAND mode.

Say	What happens
"Arm loc"	Selects V/L mode.
"Altitude _ pull or Flight level ___ pull"	Sets commanded altitude and pulls altitude knob.
"Altitude pull or Flight level pull"	Pulls altitude knob.

Flight Controls Check

Say	What happens
"Flight controls check"	Starts the flight controls flow: Up, Down, Left, Right, Rudder Left, Rudder Right

Launching Flows by Voice

Say	Flow launched
"Clear left" or "Left side clear"	Clear Left flow
"Runway entry procedure" or "Clear to line up"	Before Takeoff flow
"Before start procedure"	Before Start flow

Launching Checklists by Voice

Say	Checklist launched
"Before start checklist to the line"	Before start to the line
"Before start checklist below the line"	Before start below the line
"After start checklist"	After Start
"Before takeoff checklist to the line"	Before takeoff to the line
"Before takeoff checklist below the line"	Before takeoff below the line
"After takeoff climb checklist to the line"	Climb to the line
"After takeoff climb checklist below the line"	Climb below the line
"Approach checklist"	Approach
"Landing checklist"	Landing
"After landing checklist"	After Landing
"Parking checklist"	Parking
"Secure aircraft checklist"	Secure Aircraft
"Stop checklist" or "Abort checklist" or "Cancel checklist"	Aborts the active checklist

Tips & Troubleshooting

The FO isn't hearing me

- Check that your microphone is selected and working.
- Adjust the **Voice Sensitivity** setting.

The FO keeps repeating the challenge

- Your response didn't match the expected phrase. Listen to the challenge and use one of the phrases listed in this manual (voice matching can be tuned in settings).
- If a physical switch must be set first (e.g., parking brake), set it in the cockpit before responding.

How do I stop a checklist mid-way?

- Say "**Stop checklist**", "**Abort checklist**", or "**Cancel checklist**" at any time.

Can I run flows and checklists manually without voice?

- Yes. Both can be triggered from the **Flows** and **Checklist** panels in the app UI.